

Connectopia: A Modular Board Game Simulating Digital Interfaces to Assist Refugee and Asylum Seeker Integration

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ABSTRACT

Interactive technologies play an increasingly vital role in the integration processes for refugees and asylum seekers, but a lack of familiarity makes their use challenging and stressful. We aimed to create an artefact that could help migrant centres provide better digital-skills education to refugees while using playful and social design elements. Interviews with staff members at UK migrant centres were processed in a thematic analysis, revealing a high demand for help with integration-related digital tasks among refugees. This led to our creation of Connectopia, a two-player, competitive boardgame that uses modular components, an internal microcontroller and multimodal feedback to playfully emulate interfaces such as government portals and email. We created a low-fidelity prototype, which allowed us to perform user testing and a heuristic evaluation. The final design expands on the original by including multimodal feedback (haptic, visual, and auditory) to communicate success or failure at tasks. Connectopia can be expanded to simulate a variety of interfaces and provide playful education in refugee and migrant centres, in both social and structured educational contexts.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

Connectopia, Refugee Integration, Modular Board Game, Digital Literacy, Asylum Seeker Assistance, Digital Interface Simulation, Interactive Learning Tool

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1 INTRODUCTION

Refugees and asylum seekers may only constitute 0.5% of the United Kingdom's population[20], yet the number of asylum seeker applications is the largest since its peak in 2002[17]. The immigration and integration processes are two of many challenges facing refugees arriving in the UK, with many parts of this bureaucracy relying on unfamiliar technologies. Although there is a notable reliance on smartphones throughout refugees' flight journeys, many refugees experience linguistic, financial or cultural barriers hindering their use of the technologies required for integration[2]. As a result, many refugees lack the skills needed to approach these processes with digital literacy. Digital literacy, defined as "the ability to use information and communication technologies to find, evaluate, create, and communicate information, requiring both cognitive and technical skills"[4]. Research has demonstrated digital literacy to be crucial during the integration process between host societies and refugees[15].

Human-computer interaction (HCI) research has assessed which technologies refugees and asylum seekers rely on prior to their arrival[3] and during their resettlement[13], as well as for their mental health [12]. However, it remains unclear how HCI research can be best utilised to assist in their post-arrival integration. To address this research gap, we aimed to discover real-world problems refugees and asylum seekers might experience during their integration process, and consequently create a technologically relevant solution.

The 10th United Nations Sustainable Development Goal targets the reduction of inequality within and among countries[18]. Similarly, the 16th Goal focuses on the promotion of peaceful and inclusive societies for sustainable development, the provision of access to justice for everyone and building effective, accountable, and inclusive institutions at all levels[19]. Both goals are closely linked to refugees and asylum seekers who arrive in the UK in hopes of a better future and smooth integration into society.

We aimed to create an interactive technology assisting refugees and asylum seekers in their integration process. After conducting a literature review and interviews with refugee centre workers, we identified a lack of training and technophobia as impediments to refugees' digital integration. Our design therefore focused on increasing digital skills and reducing technology-related anxiety in refugees and asylum seekers. After utilizing a user-centred design approach, we arrived at a modular educational artefact of a board

game simulating the experience of using an email interface aiming to foster a low-stress, no-stakes environment. Consequently, our research builds upon previous work in several ways and makes the following new contributions:

- The artefact could help refugees and asylum seekers feel more comfortable using technology.
- The artefact demonstrates a creative, communal yet educational approach to bridging the technological needs of refugees and asylum seekers, as well as reducing work for staff at refugee centres.
- An adapted artefact design could be effective in alternative settings e.g. children expanding technological skills within educational settings and elderly individuals with technophobia.

2 LITERATURE REVIEW

Our process began by conducting a literature review to explore techniques for technology-related fear reduction. This informed the design of our interactive technology, emphasizing the importance of modularity, hands-on elements and social interactions.

Batool et al. conducted a study with female health workers in Pakistan with limited technological exposure [6]. They aimed to reduce technological fear pertaining to collecting and processing health data. Even governmental financial incentives were insufficient for the transition from manual to digital technological health data implementations. This was largely attributed to poor application interface design. User-friendliness is thus essential when creating and implementing a novel artefact aimed at reducing technological fear. Similarly, Atkinson et al. deduced that interactively showing individuals how to interact with a selected technological tool is more beneficial than verbal instructions [5]. This increased their self-efficacy, allowing them to feel more in control of their technology usage. Moreover, goal setting, self-regulation of progress and the ability to adapt their behaviour accordingly were found to be beneficial towards lowering technological anxiety. Our artefact therefore sought to increase users' perceived self-efficacy when using unfamiliar technologies through interactive, rather than verbal instructions.

Furthermore, in a study utilising LEGO blocks and storytelling with young Syrian refugees, researchers found the hands-on element of the activity particularly engaging and enjoyable, encouraging a hands-on approach for the interaction with our study's artefact[10]. A board game, much like LEGO blocks, is a hands-on, fun and engaging activity. Additionally, social interactions relating to technology usage resulted in diminished technology-related anxiety [11]. Many refugees rely on refugee centres to get help with integration; given the social nature of many of these centres, a playful and social intervention may be most appropriate for addressing digital skills.

3 USER RESEARCH

Due to ethical clearance constraints received for this research, we were unable to gather data directly from refugees and migrants. This led us to conduct semi-structured interviews with staff at the centres. Our interviews aimed to enhance our understanding of digital skills training methods in refugee centres. Based on the

literature, we focused our interviews on the topics of technophobia, digital skills training tools, and the technologies required for integration. This allowed us to better understand the problem space.

3.1 Participants

We recruited six individuals working at refugee centres with ages ranging between 25 and 59 years old. Participants held a range of roles within their respective centres, with many holding directorial or managerial positions, allowing for a variety of perspectives. The migrant centres included in the sample were in urban areas in the UK. Interviews were conducted by two alternating researchers over Microsoft Teams.

3.2 Analysis and results

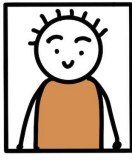
All six interview transcripts were imported into NVivo, where a bottom-up thematic analysis was conducted[8]. With the objective of identifying a clear problem space present in refugee and migrant centres, participant statements were openly coded. Following several rounds of analysis, we refined the emerging codes into the following three key themes which informed the rest of our design process:

- (1) Refugees are required to use unfamiliar technologies to integrate. Communications from the UK Home Office arriving via email with links to government portal websites are highly important, as individuals' timely responses to them are required for the maintenance of their immigration status. Many basic needs, such as accommodation benefits, also rely on email communications.
- (2) Interacting with these unfamiliar technologies is stressful for many refugees. Refugees who are unfamiliar with these emails never develop the habit of routinely checking their inboxes and spam, missing important communications. Added pressure from the legal consequences of failing to respond exacerbates the technophobic stress that many refugees face.
- (3) Many refugees do not trust themselves to handle high-stakes tasks online and rely on help from workers at centres to do so. This suggests that many of the current digital skills training courses at these centres are not supplying long-term increases in self-efficacy. Some participants noted that this behaviour was particularly frequent amongst older individuals at their centres.

Beyond these core themes, many of the participants also described their centres as community- and socialisation-driven; though this was not a key element of all centres described in the interviews, it was significant enough to merit exploration in our design process.

3.3 Personas

Based on the information from the semi-structured interviews, we created two personas (Figures 1 and 2 below) that reflected many of the individuals who rely on refugee centres for digital skills training. Highlighting the core themes of trust, tech-related negative emotions and a lack of familiarity with certain digital technologies, the personas represent two different but overlapping experiences found in the refugee community.



Name: Josh
Age: 24
Time in UK: <1 year
Goals: Gain professional tech skills for job
Digital Empowerment: 5/10
English Proficiency: 2/10

Figure 1: Persona representing a refugee/asylum seeker who has only been in the UK for a few months.



Name: Linda
Age: 61
Time in UK: 6 years
Goals: gain familiarity with technology to stay connected with loved ones
Digital Empowerment: 2/10
English Proficiency: 6/10

Figure 2: Persona representing a refugee/asylum seeker who has been in the UK for several years.

4 LOW-FIDELITY PROTOTYPE

4.1 Overview of Prototype

Our research and iterative design process resulted in a low-fidelity prototype of an interactive board game simulating an email interface (Figure 3). Expanding on findings demonstrating the effectiveness of a hands-on approach, we emphasise tangible modularity in our design [10]. We achieved this by dividing the simulated interface into various physical components. We used cardboard to represent the modular pegboard and different detachable elements of an email interface – namely, an "inbox folder," "spam folder," "delete," "create new email," "flag," and "mark as not spam". These modular components were attached to the board with blue tack. We also incorporated audio (different sounds for correct and incorrect actions) and haptic (vibration) feedback and designed instructions to clearly outline the goals of each task [5].



Figure 3: Image of the first version of our low-fidelity prototype.

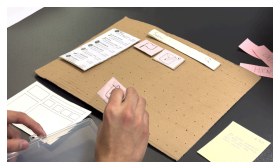


Figure 4: Image of the iterated version of low-fidelity prototype with player completing the task.

4.2 Findings from pilot testing our prototype

We conducted a pilot test with two HCI students to evaluate the basic functionality and concept of our design. We found that it was not clear how the modular elements should be placed on the board; to fix this, we poked holes in the board to more closely represent a modular "pegboard" and help guide users when placing the elements. We also added hint options, as users found it difficult

to complete certain tasks. We decided to build on our prototype to create a two-player board game with a scoreboard which tracks users' points for each task, hoping that this will also increase long-term engagement. All of these changes are shown in Figure 4 above.

4.3 User testing procedure with iterated prototype

To further evaluate the functionality of our prototype, we recruited five participants to take part in wizard-of-oz think-aloud sessions. These sessions simulated a two-player interactive experience of our game. One team member drew a task card (example shown in Figure 4) and facilitated the task's completion for participants, including the provision of hints, handing over components when needed, and simulating audio and haptic feedback. Participants completed those tasks by interacting with the interface to stick, move or click on components. Participants were subsequently asked to fill out a follow-up survey aiming to evaluate the functionality, feedback mechanisms and overall enjoyment of our game.

4.4 User testing insights

All participants found the activity enjoyable, particularly the interactive nature of the game, the use of haptic and auditory cues and the availability of hints. However, some participants found the board game's goals unclear. This indicated a need for better onboarding. Additionally, confusion arose regarding navigation elements, particularly when tasks required participants to transition from the inbox to spam.

4.5 Heuristic evaluation of prototype

An evaluation of the prototype was conducted according to the Nielsen-Norman Group's usability heuristics[14].

Heuristic	Violation	Suggested Improvements
Visibility of system status	Players do not receive any feedback on how close they are to completing each game task.	A progress bar can be included to visualise the current state of the game, including current player scores, game turn, and game task completion status.
Consistency and standards	Some of the game pieces used icons while others used words to communicate their function. This inconsistency may cause confusion for players.	Where possible, game pieces should rely fully on symbols rather than words, as well as providing visible and tactile feedback when touching clickable game pieces.
Recognition rather than recall	One task required users to find an email in the inbox according to a sender's name.	Allow task cards to be easily visible throughout the task progress.

Table 1: Notable heuristic violations in our low-fidelity prototype and potential improvements considered for our final design.

5 FINAL DESIGN

5.1 Introducing Connectopia

Incorporating participants' feedback, we aim to transform our low-fidelity prototype into a more technologically advanced learning tool, titled "Connectopia" (shown in Figure 5 below). The game uses connectable LCD screens and pressable buttons, both of which have magnets on the back and can snap into any place on the MGB. Using LCDs allows Connectopia to simulate a wider variety of interfaces, as the screens will be able to display various possible elements of an interface. For a simulated email task, the email sidebar and email content use LCD screens to enable reusability and versatility of future content; to achieve this, they will be coded to display predetermined inboxes (i.e. inbox, spam etc.) and emails (e.g. simulated responses from the NHS). The buttons allow users to "delete," "create new," "flag," and "mark as not spam" the simulated emails; these buttons use symbols to convey their meaning.

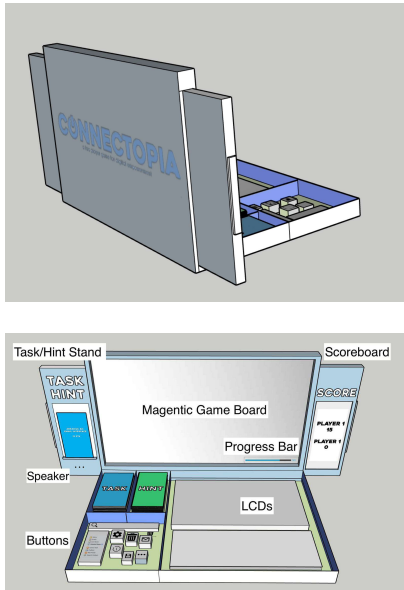


Figure 5: 3D rendering of Connectopia with labelled features and components

Additionally, users will need to place drawn task and hint cards onto the task stand. The cards will be embedded with a unique Radio Frequency Identification (RFID) tag. The RFID reader will recognise the tag and adjust the feedback of the board game to match the required task/hint card.

5.2 Connectopia's technical functionality

A key aspect of our design is the creation of a robust software program. This program will be executed on an embedded system within the gaming board: a small, low-power microcontroller or microcomputer, e.g. Arduino or a Raspberry Pi. It will continuously monitor signals from the sensors and RFID readers installed on the

board; therefore, when the user interacts with components, this information will be processed by the software to provide visual, audio and haptic feedback. Research has demonstrated that multimodal feedback can enhance the amount of information transferred when performing tasks[1]. We incorporated three modes of feedback in hopes of maximising the information transfer. Visual (coloured LEDs) and haptic feedback from actuators will be provided alongside auditory signals from small built-in speakers, notifying the user whether their interactions are correct or incorrect. Additionally, the progress bar and scoreboard will be automatically updated. Players can attempt to complete a task by moving the relevant components onto and around the MGB and interacting with them. Hint cards are readily available if needed.

6 DISCUSSION

Our thematic analysis highlighted technological self-efficacy and staffing shortages as key barriers to effective digital skills learning within refugee and asylum centres in the UK. Our two-player game with varying difficulty levels emphasises user-friendliness and social interactions, both of which have been shown to support technological self-efficacy and enable implementation without interference from staff members[6][11]. We tailored our design to specifically simulate email interfaces, considering that self-efficacy with email use is particularly weak among refugees and may be restricting their integration. Connectopia's modularity allows it to be adapted to interfaces beyond email, such as government portals or job-hunting websites. Future research could adapt our tool to be applied within other settings where a similar interactive and intuitive approach to the design of digital learning tools is encouraged.

6.1 Study limitations

Although findings from our usability testing demonstrate effective functionality, participants were students who are fluent in English and proficient with email, making them less representative of typical refugees and asylum seekers who may have significant language barriers[9]. It is not clear whether our tool would improve refugees' self-efficacy with emails and, if so, whether this would generalise to other types of interfaces, such as government portals and job application sites. To address this, it is highly recommended that future research include refugees and key staff members as part of the design process.

6.2 Future directions

There is little understanding in the literature regarding what technological skills are most helpful to learn during different stages of refugees' integration process[7]. Conducting a longitudinal study with thorough observations from refugee centres could prove beneficial. This could help determine whether our tool should be implemented in existing digital skills training courses or as games used communally within centres. Potocky's review highlights the gap in research regarding the different factors that influence refugee's trust in different digital technologies[15]. Better understanding these factors could help inform the design of future iterations whilst also providing a basis for measuring changes in refugees' trust in technology in the long-term. Consequently, it may be helpful to approach technophobia as a form of anxiety. Sherrill

et al. successfully adapted a theoretical model of exposure therapy to explore whether their telehealth tool effectively reduced technophobia among clinicians[16]. This approach could be applied to maximise the impact of Connectopia within refugee centres and across other settings where technophobia is prominent, e.g., with elderly individuals who are fearful of modern technological developments[11].

7 CONCLUSION

This paper reports our research and design process aimed at addressing technophobia and digital illiteracy in refugee and asylum seeker communities in the UK. We identify this issue as falling under the UN Sustainable Development Goals 10 and 16. A literature review and interview study with stakeholders at migrant centres revealed that many refugees experience anxiety when required to use certain digital services (ie. email and official websites) and rely on help from centre staff. These insights, coupled with an understanding of migrant centres as social and learning spaces, led us to the creation of Connectopia. Our design intervention is a board game aimed at reducing technophobia and increasing the digital skills of our users while promoting communal learning in a social setting. Longitudinal research is needed to measure the long-term impact of Connectopia on users' technophobia and digital skills, and further user research is also needed to fine-tune the gameplay mechanics. Connectopia takes a novel approach to reducing some of the stressors associated with refugee integration and future iterations should focus on expanding the scope and impact of similar interventions.

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